Review: STONE: A Spatio-temporal OOD Learning Framework Kills Both Spatial and Temporal Shifts

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1 Abstract

Traffic prediction is a crucial task in the Intelligent Transportation System (ITS), receiving significant attention from both industry and academia. Numerous spatio-temporal graph convolutional networks have emerged for traffic prediction and achieved remarkable success. However, these models have limitations in terms of generalization and scalability when dealing with Out-of-Distribution (OOD) graph data with both structural and temporal shifts. To tackle the challenges of spatio-temporal shift, we propose a framework called STONE by learning invariable node dependencies, which achieve stable performance in variable environments. STONE initially employs gated-transformers to extract spatial and temporal semantic graphs. These two kinds of graphs represent spatial and temporal dependencies, respectively. Then we design three techniques to address spatio-temporal shifts. Firstly, we introduce a Fréchet embedding method that is insensitive to structural shifts, and this embedding space can integrate loose position dependencies of nodes within the graph. Secondly, we propose a graph intervention mechanism to generate multiple variant environments by perturbing two kinds of semantic graphs without any data augmentations, and STONE can explore invariant node representation from environments. Finally, we further introduce an explore-toextrapolate risk objective to enhance the variety of generated environments. We conduct experiments on multiple traffic datasets, and the results demonstrate that our proposed model exhibits competitive performance in terms of generalization and scalability.

1.1 Keywords

Spatio-temporal data mining, Out-of-distribution generalization, Traffic prediction, Causal graph learning

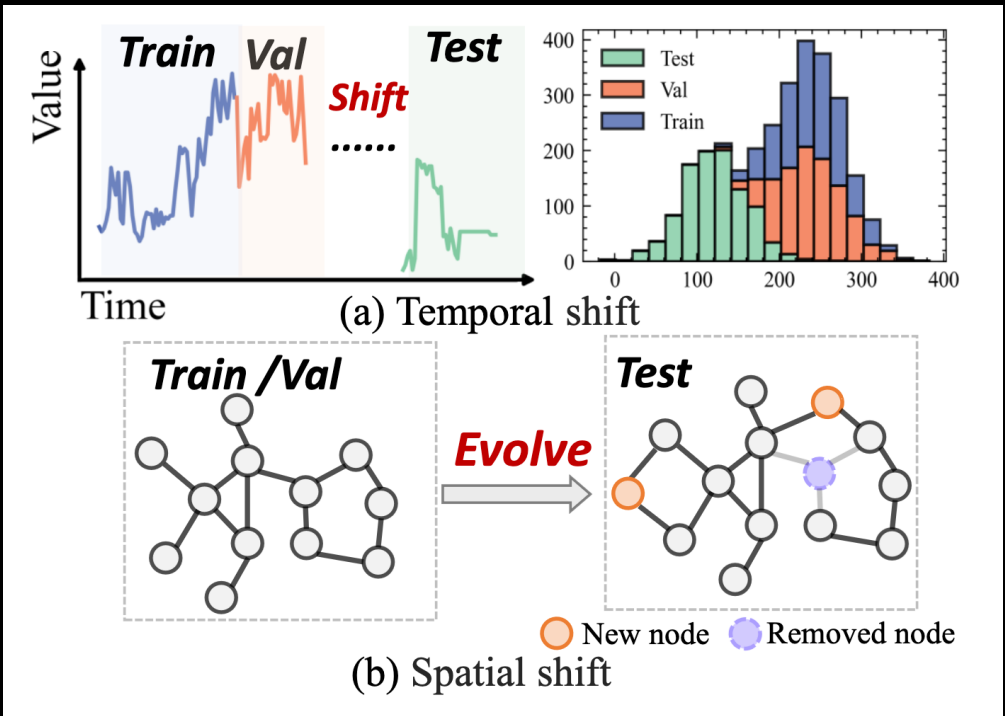
2 Introduction

- GPS-enabled mobile data is a valuable asset in realm of urban computing
- Current traffic flow prediction uses data in graph-structured data in spatio-temporal graphs (STG)

- STG convolutional networks used to make spatio-temporal features and make predictions
- For graph $\mathbf{G}$ and training data $\mathbf{X}$ in the environment $e : \{\mathbf{X}, \mathbf{G}\}$, STG prediction learnings function $\mathcal{F}$ that predicts target y given input x:

$$\min_{\mathcal{F}} \mathbb{E}_{(x,y)\sim P(x,y|e)}[\mathcal{L}(\mathcal{F}(\mathbf{x}), y)|e]$$

- Models are succesful bur heavily depend on IID assumption, not true in real world where STG may change over time, posing issues with OOD scenarios
- Recent work in OOD spatio-temporal learning focusing on temporal shift but not spatial



(a) Temporal shift refers to change in temporal data distribution, $P(\mathbf{X}_{te}) \neq P(\mathbf{X}_{tr})$

(b) Spatial shift reffers to the evolution of underlying graph structure. Example is new or removed traffic sensor nodes in network

Formally

References

[1] Binwu Wang, Jiaming Ma, Pengkun Wang, Xu Wang, Yudong Zhang, Zhengyang Zhou, and Yang Wang. STONE: A Spatio-temporal OOD Learning Framework Kills Both Spatial and Temporal Shifts. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '24, pages 2948–2959, New York, NY, USA, August 2024. Association for Computing Machinery. ISBN 979-8-4007-0490-1. doi: 10.1145/3637528.3671680. URL `https://dl.acm.org/doi/10.1145/3637528.3671680`.